

- 4.1 For an NMOS differential pair with a common-mode voltage V_{CM} applied, as shown in Fig. 4.1, let $V_{DD} = 2V$, $\mu_n C_{ox} = 0.4 \text{ mA/V}^2$, $(W/L)_{1,2} = 10$, $V_{TH} = 0.4 \text{ V}$, $I_{SS} = 0.16 \text{ mA}$, $R_D = 5 \text{ k}\Omega$, and neglect channel-length modulation and body effect.
- Find V_{OD} and V_{GS} for each transistor.
 - For $V_{CM} = 1V$, find V_P , I_{D1} , I_{D2} , V_{OUT1} , and V_{OUT2} .
 - Repeat (b) for $V_{CM} = 1.4 \text{ V}$.
 - Repeat (b) for $V_{CM} = 0.9 \text{ V}$.
 - What is the highest value of V_{CM} for which M_1 and M_2 remain in saturation?
 - If current source I_{SS} requires a minimum voltage of 0.2 V to operate properly, what is the lowest value allowed for V_P and hence for V_{CM} ?
 - Find the small-signal differential gain under the condition of (b).

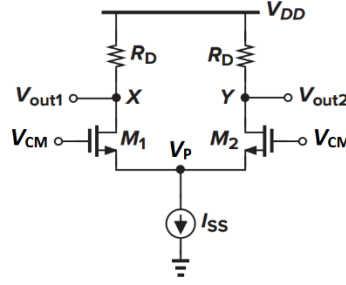


Fig. 4.1

- 4.2 Consider the differential amplifier specified in Problem 4.1 with gate voltage of M_2 , V_{G2} , fixed to $V_{CM} = 1V$, and V_{G1} is set to $V_{CM} + v_{id}$. Let v_{id} be adjusted to the value that causes $I_{D1} = 0.09 \text{ mA}$ and $I_{D2} = 0.07 \text{ mA}$.
- Find the corresponding values of V_{GS2} , V_P , V_{GS1} , and hence v_{id} .
 - What is the difference output voltage $v_{od} = V_{out2} - V_{out1}$?
 - What is the voltage gain v_{od}/v_{id} ? Explain the reason why the gain obtained here is slightly different with that obtained in 4.1 (g).
 - What value of v_{id} results in $I_{D1} = 0.07 \text{ mA}$ and $I_{D2} = 0.09 \text{ mA}$?
 - What is the maximum range of v_{id} that ensures either I_{D1} or I_{D2} be larger than 0.

- 4.3 Fig. 4.3 shows a MOS differential amplifier with the drain resistors R_D implemented using diode-connected PMOS transistors, M_3 and M_4 . Let M_1 and M_2 be matched, and M_3 and M_4 be matched. Neglect body effect.
- Draw the differential half-circuit and use it to derive an expression for A_{DM} in terms of $g_{m1,2}$, $g_{m3,4}$, $r_{o1,2}$, and $r_{o3,4}$.
 - Neglecting the effect of the output resistances r_o , find A_{DM} in terms of $\mu_n C_{ox}$, $\mu_p C_{ox}$, $(W/L)_{1,2}$ and $(W/L)_{3,4}$.
 - With the condition of (b), if $\mu_n = 4\mu_p$ and all four transistors have the same channel length, find $(W_{1,2}/W_{3,4})$ that results in $A_{DM} = 10 \text{ V/V}$.
 - Find the maximum $V_{in,CM}$ that ensures saturation of M_1 and M_2 in terms of V_{DD} , I_{SS} , $\mu_p C_{ox}$, $(W/L)_{3,4}$, V_{THP} and V_{THN} .

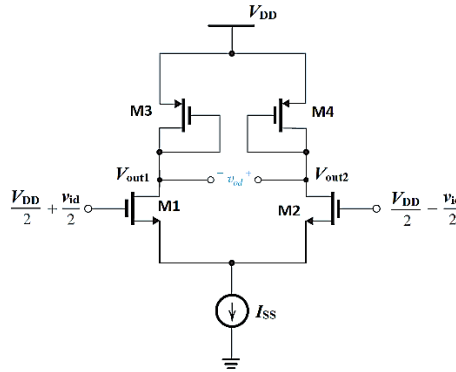


Fig. 4.3

4.4 For the differential amplifier shown in Fig. 4.4, neglect the channel length modulation effect.

- Draw the differential half-circuit for the amplifier and use it to derive an expression for the differential gain $A_{DM} \equiv v_{od}/v_{id}$ in terms of g_m , R_D , and R_S .
- What is the gain with $R_S = 0$?
- What is the value of R_S (in terms of $1/g_m$) that reduces the gain to half of value at $R_S = 0$?
- Derive the maximum range of v_{id} (in terms of g_m , I_{SS} , R_S) that ensures either I_{D1} or I_{D2} be larger than 0.

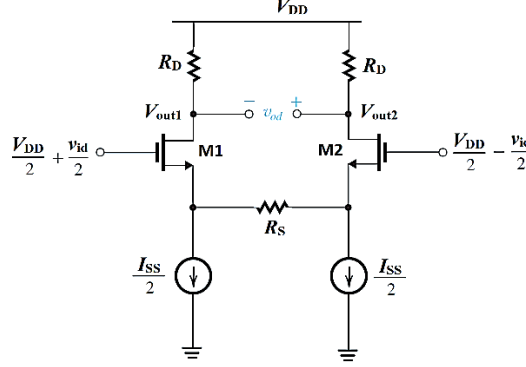


Fig. 4.4

4.5 A design error has resulted in a gross mismatch in the circuit of Fig. 4.5. Specifically, M2 has twice the W/L ratio of M1. Neglect channel length modulation effect and body effect. If v_{id} is a small AC signal, find:

- I_{D1} and I_{D2} for $v_{id}=0$.
- The offset voltage, $v_{od,offset}$, at the output nodes. (Hint: $v_{od,offset} = V_{out1} - V_{out2}$ when $v_{id}=0$.)
- V_{OD} for each of M1 and M2 in terms of I_{SS} , W/L and $\mu_n C_{ox}$.
- The differential gain A_{DM} in terms of R_D , I_{SS} , and V_{OD} . (Note: you cannot use the half-circuit method as the circuit is asymmetrical)

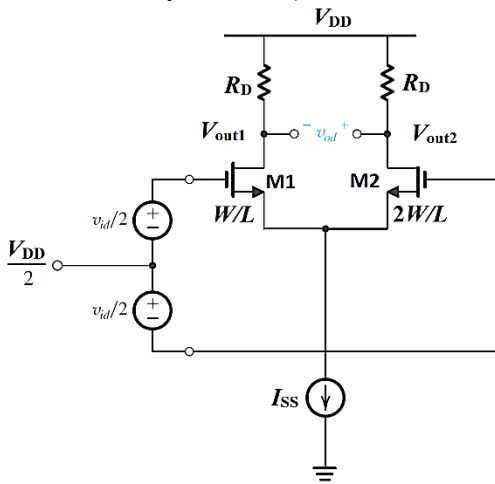


Fig. 4.5

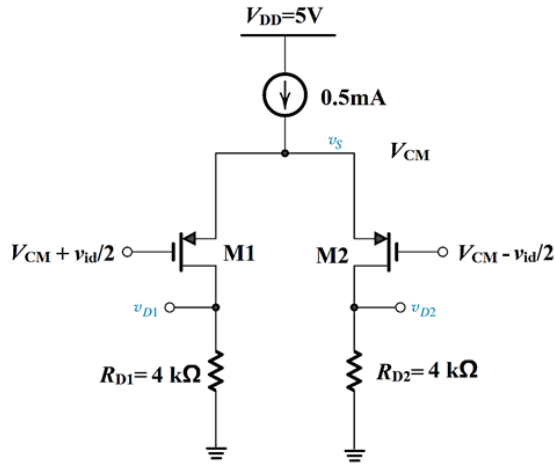


Fig. 4.6

4.6 For the differential amplifier shown in Fig. 4.6, let M1 and M2 have $\mu_p C_{ox} = 0.4 \text{ mA/V}^2$, $W/L=10$, and assume that the bias current source has an output resistance of $30 \text{ k}\Omega$. Neglect channel length modulation effect and body effect

- Find V_{OD} , g_m , the differential-mode gain A_{DM} and the common-mode gain A_{CM} .
- If the drain resistances are known to have a mismatch of 2%, e.g. $R_{D1}=4\text{k}\Omega*(1+1\%)$ and $R_{D2}=4\text{k}\Omega*(1-1\%)$, find the A_{CM-DM} , and the CMRR in dB (i.e. $\text{CMRR}_{\text{dB}}=20\log_{10}(A_{DM}/A_{CM-DM})$).
- On the basis of the condition given in (b), if the width of the M1 and M2 have a mismatch of 2% further, find A_{CM-DM} , and CMRR (in dB) again.